

Dynamic Classification of Fragile and Antifragile Response Legs

A causal, exhaustive, two-leg implementation of the signed variance-response rule

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Abstract

This article specifies and audits a dynamic classifier that assigns every eligible ticker to exactly one of two response legs after each market close. The estimator uses the MASS signed variance-response relation $E_{i,s} = \alpha_{i,t} - \beta_{i,t}V_{i,s} + \varepsilon_{i,s}$ on the latest 252 available regression observations. A positive coefficient defines the fragile response leg R^+ ; a negative coefficient defines the antifragile response leg R^- . The legs are mutually exclusive, jointly exhaustive for valid observations, and never replaced by a third “neutral” regime. Statistical confidence is exported separately from leg membership.

The empirical run 20260711T051344.464872Z contains 419 tickers with at least ten years of Yahoo Finance price history. At the terminal signal date, 267 tickers are in R^+ and 152 are in R^- . An independent coefficient reconstruction reproduces all 419 signs, the mapping between coefficient sign and leg has zero discrepancies, and the full daily history contains zero sign-leg inconsistencies across 3,439,101 classified rows. These are software-classification accuracy results. They are distinct from robustness diagnostics: 120 of 419 current classifications pass all six additional inference and stability controls. Every ticker nevertheless remains in one and only one leg.

The balanced panel contains 2,015 signal dates from 3 July 2018 through 10 July 2026. It contains 16,913 leg changes: 8,449 transitions from R^+ to R^- and 8,464 transitions from R^- to R^+ . Current option chains are timestamped and causally aligned with OHLC data, but options do not enter the R^+/R^- classification. The output is a state classifier and an input to later pair construction, not a return-prediction model or an investment recommendation.

Keywords: fragility; antifragility; signed variance response; R^+ ; R^- ; rolling regression; event time; HAC covariance; block bootstrap; no look-ahead.

JEL codes: C22; C58; G11; G17.

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1 Objective and scope

The objective is narrow and operational: construct the two legs required by a fragile-antifragile pair system. For every eligible asset and every date with a valid estimate, the classifier must answer one question: does the asset's locally adjusted performance decrease or increase as realized variance rises? The answer creates the two disjoint legs

$$\mathcal{U}_t = \mathcal{R}_t^+ \dot{\cup} \mathcal{R}_t^-, \quad \mathcal{R}_t^+ \cap \mathcal{R}_t^- = \emptyset,$$

where $\mathcal{U}_t$ is the valid universe at date t . A ticker cannot belong to both legs on the same date. There is no multi-regime assignment and no third classification state.

The classifier is the first layer of a larger construction. It identifies the eligible growth/fragility carrier R^+ and the eligible pressure/antifragility carrier R^- . Pair density, stress activation, density crossing, and allocation belong to the downstream MASS pair operator; they are not silently folded into the ticker classifier. This separation is essential for auditability:

$$\text{OHLC history} \longrightarrow \hat{\beta}_{i,t} \longrightarrow \{R^+, R^-\} \longrightarrow \text{pair construction.}$$

The article therefore does not claim that the sign alone proves global convexity, a causal response to an exogenous volatility shock, or profitable trading. In the terminology of [Taleb \(2012\)](#) and [Taleb and Douady \(2013\)](#), strict antifragility concerns the response of an exposure to changes in disorder and places particular weight on curvature and distributional perturbations. The implemented R^- label is the operational negative signed-response leg required by MASS. Stronger convexity and tail diagnostics are reported as separate evidence, never as a condition that removes a ticker from R^- .

1.1 What is classified, and what is not

The object being classified is the sign of a rolling response coefficient. It is not:

- a forecast of the next return;
- a permanent property of a company or ticker;
- a ranking by realized volatility level;
- an option-implied classification;
- a three-state fragile/neutral/antifragile taxonomy;
- a claim that every R^- ticker satisfies strict structural antifragility.

This distinction resolves the apparent conflict between exhaustive classification and statistical uncertainty. Leg membership answers *which sign is measured?*; confidence answers *how stable is that measured sign under specified perturbations?* Both are exported, but only the sign assigns the leg.

2 Conceptual foundations

2.1 Fragility is a response, not a volatility level

An asset with low realized volatility is not automatically antifragile, and a volatile asset is not automatically fragile. Fragility concerns the response of an outcome to a stress coordinate. In the present implementation, realized variance is the observable stress coordinate and annual excess log performance is the response. The relation is deliberately a reduced-form measurement rather than a structural model.

Let $P_{i,t} > 0$ denote the adjusted closing price of ticker i on observed market date t . Daily log return is

$$r_{i,t} = \log P_{i,t} - \log P_{i,t-1}. \quad (1)$$

The annualized realized sample variance over 252 observations is

$$V_{i,t}^{(252)} = 252 \widehat{\text{Var}}(r_{i,t-251}, \dots, r_{i,t}), \quad (2)$$

and annual excess log performance is

$$E_{i,t}^{(252)} = \log P_{i,t} - \log P_{i,t-252} - r_f, \quad r_f = 0.02. \quad (3)$$

Because the risk-free subtraction is constant within each regression window, it changes the intercept but not the OLS slope. A time-varying maturity-matched rate would be preferable for an economic interpretation of the intercept.

2.2 The MASS signed variance-response relation

The classifier follows the signed variance-response relation used by the MASS operator:

$$E_{i,s}^{(252)} = \alpha_{i,t} - \beta_{i,t} V_{i,s}^{(252)} + \varepsilon_{i,s}, \quad s \in \mathcal{W}_{i,t}, \quad (4)$$

where $\mathcal{W}_{i,t}$ contains the latest 252 valid regression rows available after the close of t . The conditional fitted response to variance is

$$\frac{\partial \hat{E}_{i,t}}{\partial V} = -\hat{\beta}_{i,t}. \quad (5)$$

Thus $\hat{\beta}_{i,t} > 0$ means variance drag: fitted excess performance falls as variance increases. This is the fragile response leg R^+ . A value $\hat{\beta}_{i,t} < 0$ reverses the response: fitted excess performance rises with variance over the observed local range. This is the antifragile response leg R^- .

The nomenclature follows the operator used for pair construction. It does not turn a linear local association into a proof of global payoff convexity. A negative coefficient may also reflect rebound dynamics, leverage effects, volatility feedback, changing risk premia, or sample composition. Those mechanisms are consistent with the broader return-volatility literature (Christie, 1982; French et al., 1987; Glosten et al., 1993; Moreira and Muir, 2017).

2.3 Two evidence layers, not three regimes

The implementation uses exactly two classification states and a separate evidence layer:

Leg R^+ if $\hat{\beta} > 0$; R^- if $\hat{\beta} < 0$. This is the complete classification rule.

Confidence

WEAK, MEDIUM, or STRONG, derived from six diagnostics. Confidence never creates or removes a leg.

Exploratory filter

The export field `taleb_validated` identifies a negative-sign candidate that also passes the implemented quadratic-curvature and tail-asymmetry conditions. It is a research filter, not a third regime and not a proof of Taleb-style structural antifragility.

3 Data, vintage, and information timing

3.1 Universe and ten-year eligibility

The source universe contains 419 unique requested tickers. All 419 pass the implementation’s terminal history rule:

- the first available adjusted price is at least ten calendar years before the run cutoff;
- the latest observation is no more than ten calendar days stale;
- coverage over the preceding decade is at least 80% of the weekday grid generated by `pandas.bdate_range`.

The last condition must be described precisely: `bdate_range` counts expected weekdays, not official exchange sessions, because it does not remove every market holiday. It is a conservative operational coverage denominator, not an exchange-calendar reconstruction.

The empirical vintage is fixed by run identifier

20260711T051344.464872Z.

The run started at 05:13:44 UTC and completed at 05:43:42 UTC on 11 July 2026. Prices extend through the observed session of 10 July 2026. Data were collected through `yfinance` version 1.4.1 (Aroussi, 2026); the numerical environment also used `pandas` 3.0.3, `NumPy` 2.4.6, `SciPy` 1.17.1, and `statsmodels` 0.14.6.

The universe is terminal and therefore subject to survivorship and selection bias. Historical breadth describes the 419 current identities projected backward. Delisted securities and former members absent from the source list are not reconstructed.

3.2 OHLC and option data

Yahoo Finance provides open, high, low, close, adjusted close, volume, dividends, splits, and provider timestamps. The classifier uses adjusted closes. Every OHLC row stores its market event date, estimated close timestamp, availability timestamp, fetch timestamp, and run identifier.

The same run stored 301,598 currently listed option contracts across all expirations returned by Yahoo. Option download status was OK for 385 tickers, PARTIAL for 32, and NO_OPTIONS for 2. The run-level status is therefore PARTIAL; this status describes optional chain availability, not a failure of the OHLC classifier. All 419 OHLC classifications and exports completed.

Options are never used to estimate β , assign R^+/R^- , determine confidence, or decide pair eligibility. They are stored as timestamped snapshots for later option-layer work. The causal alignment table contains 419 ticker rows and zero negative alignment lags, so an option snapshot is never joined to an OHLC/regime observation that became available after the option information cutoff.

Yahoo exposes current option chains but not a discoverable archive of expired historical chains. Repeated runs can build a local timestamped option archive; a single run cannot recreate past chains. This limitation is why options remain outside the present classifier.

3.3 Event time and the terminal observation

Let $\mathcal{F}_{i,t}$ be the information available after the close of observed session t . The signal is adapted to that information set:

$$L_{i,t} = g(\mathcal{F}_{i,t}). \quad (6)$$

It becomes effective at the next observed session when that session exists. The export therefore stores both `date` and `signal_effective_date`. The software audit finds zero mismatches between the latter and the next observed date on every nonterminal row.

For the final observation of each ticker, the next session has not yet been downloaded. Consequently, `signal_effective_date` is empty on all 419 terminal rows. The precise convention is:

The effective date is the next observed session when it exists; for the final observation, it remains unavailable until the next download.

This convention removes direct same-session look-ahead. It does not make the relation causal, and later provider downloads may revise adjusted prices because Yahoo does not expose historical vendor vintages.

4 Estimator and unique leg assignment

4.1 Closed-form coefficient

For the latest $m = 252$ valid rows in $\mathcal{W}_{i,t}$, write $X_s = V_{i,s}^{(252)}$ and $Y_s = E_{i,s}^{(252)}$. The conventional OLS slope is

$$\hat{b}_{i,t} = \frac{\widehat{\text{Cov}}_{\mathcal{W}}(X, Y)}{\widehat{\text{Var}}_{\mathcal{W}}(X)}. \quad (7)$$

Because the model writes the slope as $-\beta$,

$$\hat{\beta}_{i,t} = -\hat{b}_{i,t} = -\frac{\widehat{\text{Cov}}_{\mathcal{W}}(V^{(252)}, E^{(252)})}{\widehat{\text{Var}}_{\mathcal{W}}(V^{(252)})}. \quad (8)$$

The intercept and coefficient of determination are

$$\hat{\alpha}_{i,t} = \bar{E}_{i,t} + \hat{\beta}_{i,t} \bar{V}_{i,t}, \quad R_{i,t}^2 = \text{Corr}_{\mathcal{W}}(V^{(252)}, E^{(252)})^2. \quad (9)$$

4.2 Exactly two legs

For every valid nonzero estimate, the rule is

$$L_{i,t} = \begin{cases} R^+, & \hat{\beta}_{i,t} > 0, \\ R^-, & \hat{\beta}_{i,t} < 0. \end{cases} \quad (10)$$

No ticker receives two labels. No confidence threshold can override Equation (10). A nonfinite estimate is an invalid calculation row, not a third economic regime; it is not admitted to the valid classified universe. In the empirical current cross-section and daily classified panel, there are no exact-zero coefficients.

The corresponding implementation labels are

$$R^+ \equiv \text{FRAGILE}, \quad R^- \equiv \text{ANTIFRAGILE}.$$

These strings are aliases for one sign rule. They do not represent separate classifiers.

4.3 Horizon estimates

The current export also contains coefficients estimated on the latest 126 and 504 regression rows. These windows are nested and the underlying annual variables overlap heavily. Their role is diagnostic: agreement across 126, 252, and 504 rows supports sign stability; disagreement lowers confidence but does not move the current 252-row estimate out of its leg.

5 Classification accuracy and software audit

5.1 Accuracy is not the same as robustness

Two questions must not be conflated:

1. **Classification accuracy of the implementation:** did the code compute the declared coefficient and map its sign to the correct, unique leg?
2. **Statistical robustness of the measured sign:** does that sign remain supported under additional dependence, resampling, horizon, deletion, outlier, and persistence checks?

The first question has a deterministic answer relative to the declared rule. The second is empirical and necessarily varies by ticker. A ticker can be correctly assigned to R^- while having weak evidence that the current negative sign is stable. Such a ticker remains in R^- with **WEAK** confidence.

Table 1: Exhaustive implementation audit for run 20260711T051344.464872Z.

Audit identity	Result
Current export rows / unique tickers	419 / 419
Duplicate current tickers	0
Ten-year eligible tickers	419 / 419
Finite, nonzero current $\hat{\beta}_{252}$	419 / 419
Independent current-sign reconstruction	419 / 419
Current sign-to-leg mismatches	0
Current regime-string-to-leg mismatches	0
Maximum absolute independent-beta error	4.58×10^{-9}
Daily rows stored / daily rows classified	3,649,858 / 3,439,101
Daily sign-to-leg mismatches	0
Nonterminal next-session date mismatches	0
Terminal rows with correctly empty effective date	419 / 419

These results support the statement that the implementation reproduces its own specified classifier exactly on the exported vintage. They do not establish that the MASS relation is a causal law or that every sign is statistically stable.

5.2 Meaning of the validation fields

Three current export fields have exact operational meanings:

classification_rule_valid	Equals one when an independent closed-form reconstruction of $\hat{\beta}_{252}$ has the same sign as the stored classification. Result: 419/419.
mass_validated	Equals one when the directional MASS sign rule is correctly applied and the ticker satisfies the ten-year eligibility filter. Result: 419/419. It does <i>not</i> mean that all six robustness diagnostics passed.
pair_eligible	Equals mass_validated in the current implementation. It indicates that the ticker can enter the later cross-leg pairing stage. Result: 419/419.

The field **classification_confidence** is independent of these three invariants. The export contains 120 **STRONG**, 148 **MEDIUM**, and 151 **WEAK** classifications. All 419 remain pair-eligible because confidence is diagnostic, not a third leg.

6 Robustness architecture and extended stress tests

6.1 The six current diagnostics

For current sign $s_{i,t} = \text{sign}(\hat{\beta}_{i,t})$, the published confidence grade uses six checks:

1. **HAC plus false-discovery control.** OLS receives Newey-West covariance with 63 lags and finite-sample correction (Newey and West, 1987; Hansen and Hodrick, 1980). Across-ticker p-values are adjusted by Benjamini-Hochberg at 5% (Benjamini and Hochberg, 1995); the HAC interval for β must exclude zero.
2. **Circular moving-block bootstrap.** Five hundred bootstrap samples use 21-session circular blocks (Künsch, 1989; Lahiri, 2003). At least 95% of bootstrap coefficients must preserve the sign and the percentile interval must exclude zero.
3. **Horizon consensus.** Coefficients on 126, 252, and 504 regression rows must all have the current sign.
4. **Delete-block stability.** Removing each nonoverlapping 21-session block in turn must preserve the sign.
5. **Winsorized stability.** Separately winsorizing variance and excess performance at 1% and 99% must preserve the sign.
6. **Recent sign persistence.** At least 80% of the latest 126 daily rolling coefficients must have the current sign.

STRONG means six passes; **MEDIUM** means four or five passes; **WEAK** means zero to three passes. These grades summarize sensitivity to the declared tests. They are not probabilities that a leg is “true.”

Table 2: Current diagnostic pass counts. Percentages use the leg-specific denominators 267 for R^+ and 152 for R^- .

Diagnostic	All (419)		R^+ (267)		R^- (152)	
	Count	%	Count	%	Count	%
HAC + FDR	234	55.8	164	61.4	70	46.1
Block bootstrap, 21	226	53.9	160	59.9	66	43.4
126/252/504 consensus	241	57.5	177	66.3	64	42.1
Delete 21-session blocks	343	81.9	221	82.8	122	80.3
Winsorize 1%	419	100.0	267	100.0	152	100.0
Recent-sign persistence	242	57.8	151	56.6	91	59.9

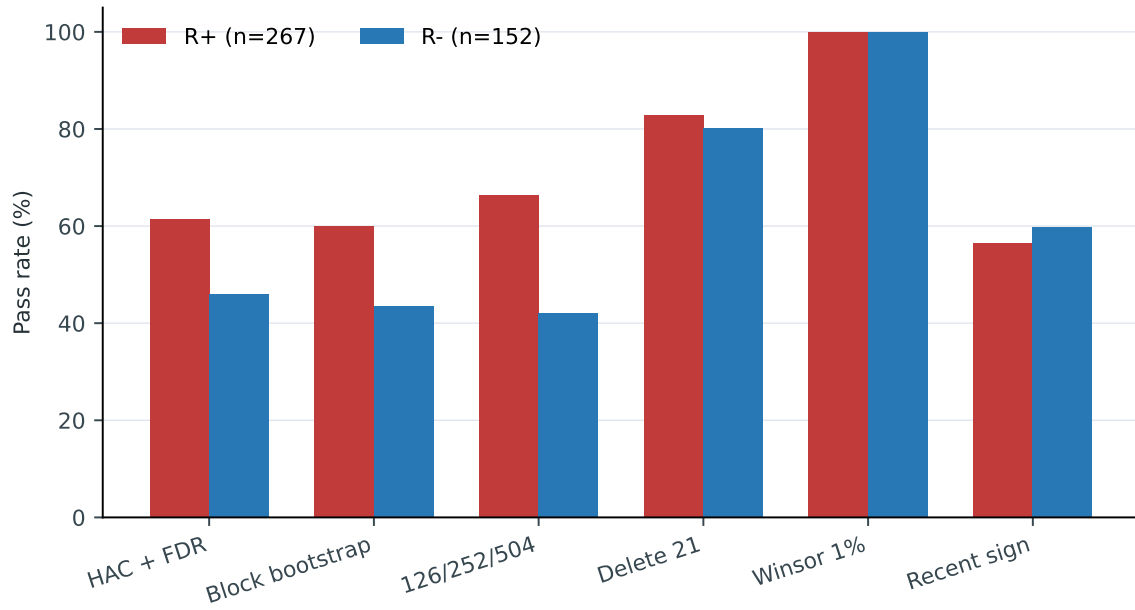


Figure 1: Pass rates for the six confidence diagnostics. The difference between a pass rate and 100% is not a classification error; it is evidence of sensitivity under that diagnostic.

Table 3: Confidence grades by unique current leg.

Leg	Weak	Medium	Strong	Total
R^+	80	97	90	267
R^-	71	51	30	152
All	151	148	120	419

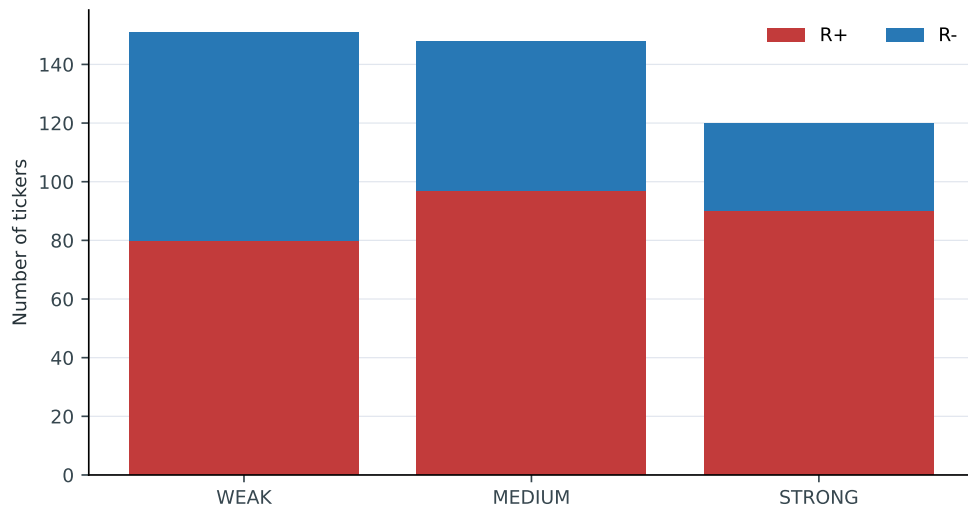


Figure 2: Confidence is an attached diagnostic and never a third regime.

6.2 Extended dependence and perturbation tests

The original 63-lag HAC and 21-session bootstrap cannot fully neutralize the 251-session overlap induced by annual variables. The audit was therefore pushed beyond the six production checks. Table 4 reports fully recalculated sensitivities using the same 419 current windows.

Table 4: Extended sign-robustness tests. Counts are diagnostic passes, not correct classifications.

Test family	Perturbation	All	R^+	R^-
HAC + FDR	21 lags	248	173	75
HAC + FDR	63 lags (production)	234	164	70
HAC + FDR	126 lags	258	179	79
HAC + FDR	251 lags	285	192	93
Circular bootstrap	21-session blocks	226	160	66
Circular bootstrap	63-session blocks	141	100	41
Circular bootstrap	126-session blocks	154	108	46
Delete-block	21-session blocks	343	221	122
Delete-block	63-session blocks	247	161	86
Delete-block	126-session blocks	196	131	65
Winsorized sign	1% / 99%	419	267	152
Winsorized sign	2.5% / 97.5%	419	267	152
Winsorized sign	5% / 95%	418	267	151
Horizon sign	126/252/504	241	177	64
Horizon sign	63/126/252/504	136	100	36

The extended results reveal genuine dependence sensitivity. Longer bootstrap blocks and larger deletion blocks reduce pass counts sharply. Adding the 63-row horizon reduces consensus from 241 to 136. Winsorization remains nearly invariant even at 5%. These results justify publishing confidence rather than claiming that all current signs are equally stable.

The HAC counts are not monotone in lag length; the 251-lag result even produces more rejections. With only 252 nominal regression rows, a 251-lag Newey-West covariance is at the boundary of what the sample can support and can underestimate uncertainty because too few effective lag products remain. It is reported as a stress diagnostic, not promoted as a superior inferential specification. Reliable long-memory inference would require a longer estimation sample, fixed-bandwidth asymptotics, nonoverlapping observations, or a predeclared alternative estimator.

6.3 Exploratory convexity and tail evidence

For negative-sign rows only, the pipeline also estimates a quadratic response in standardized variance and compares left- and right-tail semideviations between low- and high-stress observations. Sixteen of 152 current R^- tickers pass the implemented curvature and tail rules; only one is both **STRONG** and in this exploratory set. The export retains the legacy field name `taleb_validated`, but the defensible interpretation is “exploratory convexity-tail candidate.” The filter reuses data, depends on thresholds, and is not multiplicity-corrected. It cannot validate strict antifragility by itself.

7 Empirical regime results

7.1 Current cross-section

At the 10 July 2026 signal date, 267 of 419 tickers (63.72%) are in R^+ and 152 (36.28%) are in R^- . Table 5 summarizes the terminal coefficients.

Table 5: Terminal coefficient and fit summary.

Leg	Tickers	Median $\hat{\beta}_{252}$	Q1	Q3	Median R^2
R^+	267	5.119	2.522	8.593	0.295
R^-	152	-4.405	-7.463	-1.776	0.178

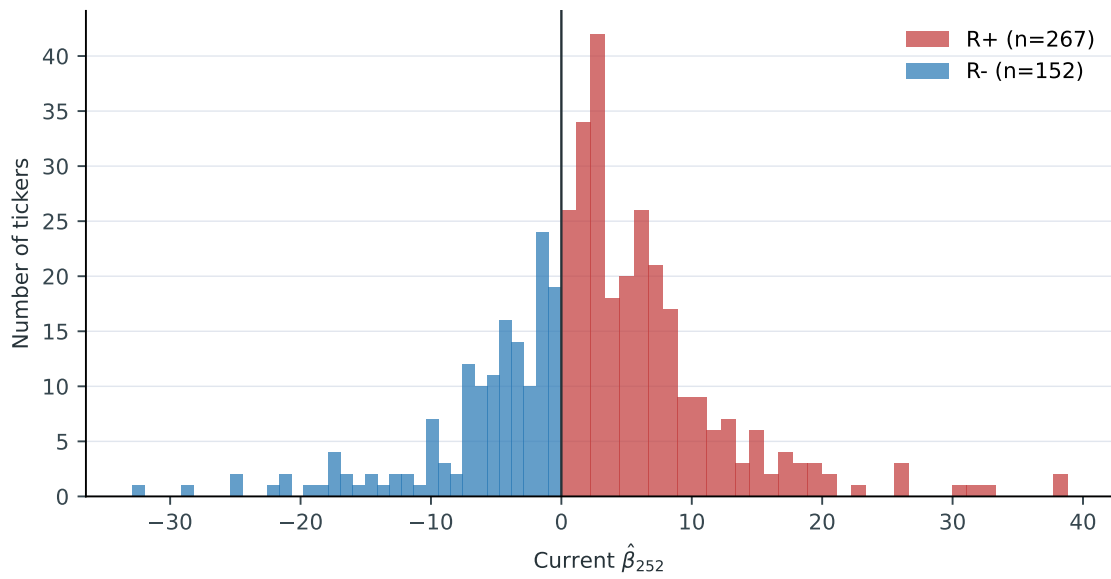


Figure 3: Distribution of the current 252-row signed variance-response coefficients. The zero boundary alone assigns the two legs.

The lower median fit in R^- and the lower pass rate for horizon consensus show that negative-response classifications are, on average, less stable under several current diagnostics. This is a statement about confidence, not about the correctness of their sign mapping.

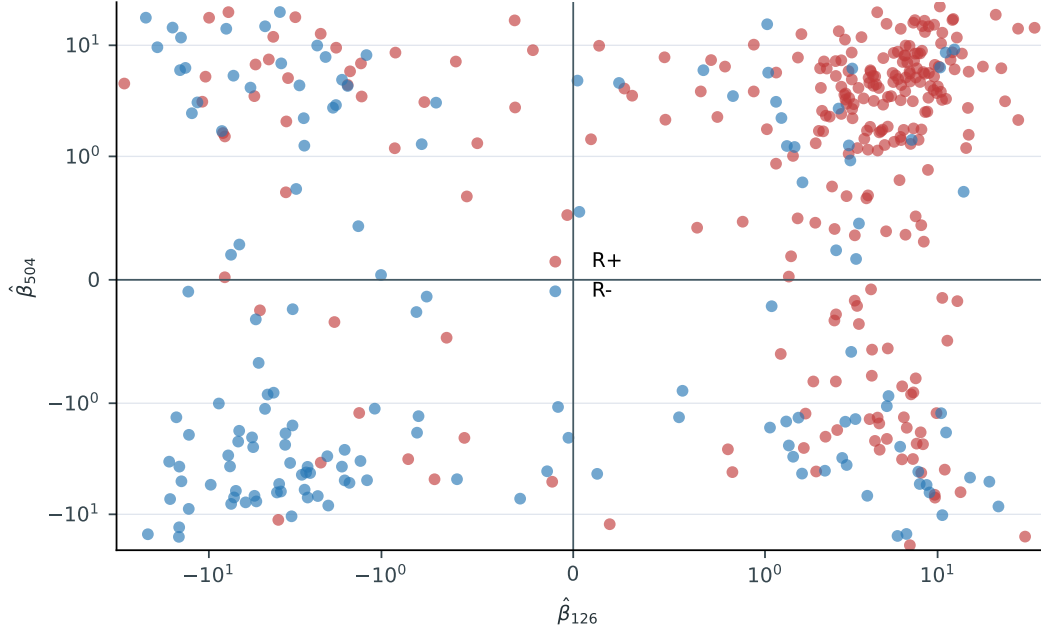


Figure 4: Current 126-row and 504-row coefficients, colored by the unique 252-row leg. Symmetric-log axes preserve the sign quadrants while displaying large coefficients.

7.2 Balanced-panel breadth

Define fragile-leg breadth as

$$B_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbf{1}\{L_{i,t} = R^+\}. \quad (11)$$

The balanced panel begins when all 419 tickers have a valid classification, on 3 July 2018. It contains 2,015 signal dates through 10 July 2026. Mean R^+ breadth is 67.67%. It reaches a maximum of 91.65% on 8 April 2020 and a minimum of 14.80% on 9 March 2022. The terminal value is 63.72%, or 267 of 419 tickers.

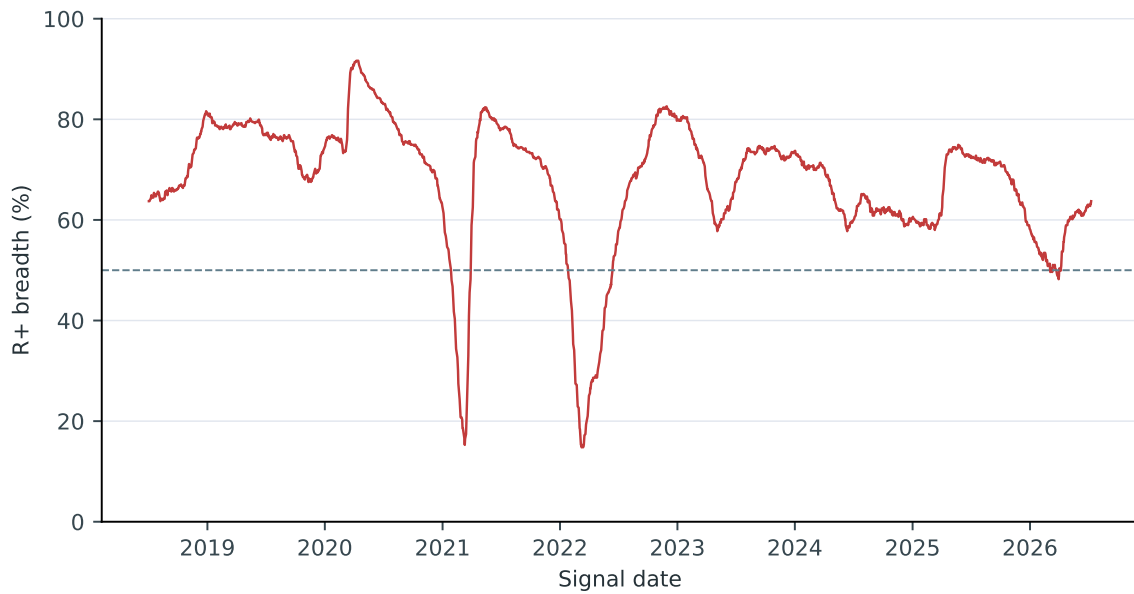


Figure 5: Balanced-panel R^+ breadth. Breadth measures how widely the adverse variance-response sign is distributed across the fixed terminal universe.

Breadth is a contemporaneous state summary. It does not forecast returns and it inherits the terminal-universe survivorship limitation.

7.3 Transitions and episode duration

The exact daily transition matrix contains 16,913 changes:

$$\begin{aligned} R^+ &\rightarrow R^- : 8,449, \\ R^- &\rightarrow R^+ : 8,464. \end{aligned}$$

Conditional on the prior leg, the one-session transition rates are 0.385% for R^+ to R^- and 0.679% for R^- to R^+ . These are the exact recomputed counts from the frozen daily export.

After excluding the first and last episode for each ticker to reduce boundary censoring, the sample contains 8,197 completed R^+ episodes and 8,297 completed R^- episodes. Median duration is 206 sessions for R^+ and 111 for R^- . Interquartile ranges are 103–320 sessions and 48–198 sessions, respectively.

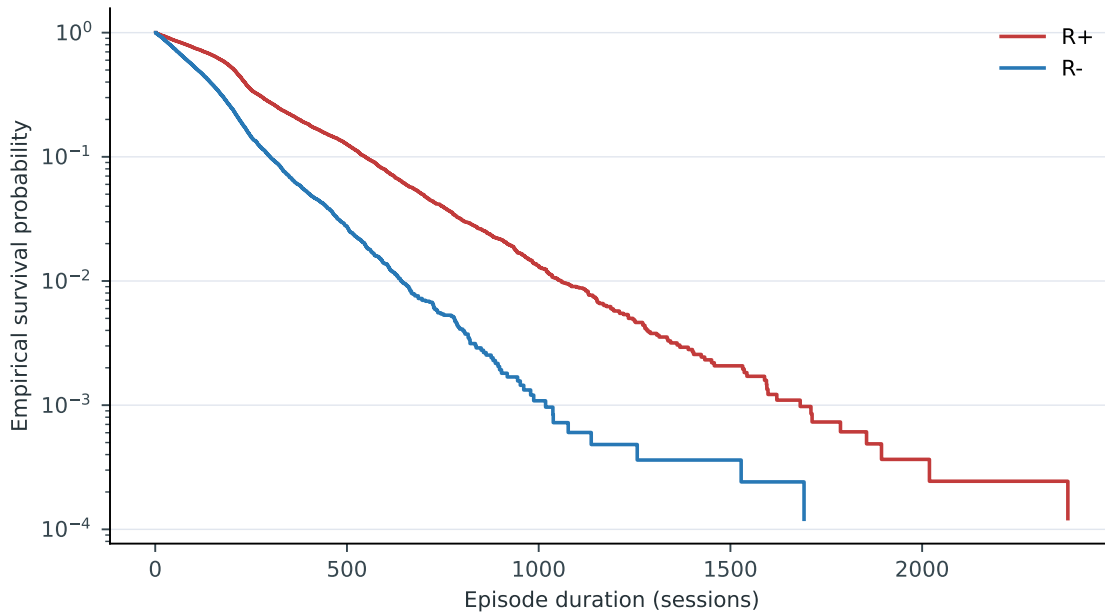


Figure 6: Empirical survival of completed leg episodes. Persistence partly reflects the 252-row estimator and the strong overlap in annual variables; it is not proof of structural permanence.

8 From classified legs to pair construction

The classifier delivers two dated sets. A later pair builder must draw one asset from $\mathcal{R}_t^+$ and one from $\mathcal{R}_t^-$:

$$\mathcal{P}_t \subseteq \mathcal{R}_t^+ \times \mathcal{R}_t^-. \quad (12)$$

This is the only admissible cross-leg quadrant. A ticker already has one unique leg at t ; pairing does not reclassify it. The current fields `mass_validated=1` and `pair_eligible=1` establish that all 419 current tickers may be offered to this next stage, subject to their unique sign.

The MASS density operator then has additional work to perform. In the downstream MASS notation, the pair builder uses the rolling positive response component $M_+(t) = \max(\beta_+(t), 0)$, the negative response component $M_-(t) = \max(-\beta_-(t), 0)$, causal stress measures, and constrained densities ρ_+ and ρ_- . Pair entry is governed by density dominance, not by the ticker classifier alone. Therefore:

- the present article validates leg construction;
- it does not claim that every possible R^+/R^- combination is an optimal pair;
- it does not use option chains to alter leg membership;
- it does not use next-day return prediction to decide the leg;
- confidence and pair-stage density diagnostics must remain separate exported fields.

9 Limitations and falsification program

Reduced-form relation. Variance and returns are jointly determined. Next-session dating prevents direct look-ahead but does not identify an exogenous variance shock. Causal claims would require instruments, natural experiments, or a structural model.

Overlap and effective sample size. Both $V^{(252)}$ and $E^{(252)}$ overlap heavily. The nominal 252-row regression contains far fewer than 252 independent annual observations. The extended bootstrap and deletion tests show that longer dependence blocks materially reduce robustness counts.

Forced two-leg partition. The project requires exactly two legs, so any finite nonzero coefficient is assigned by sign even when economically small. This is intentional. Uncertainty must be communicated through the coefficient, confidence grade, intervals, horizon signs, and persistence, not by inventing a third regime.

Constructed regressor and measurement error. Realized variance is estimated from the same daily returns that enter annual performance. Jumps, volatility clustering, and measurement error may distort the fitted slope. Intraday realized measures, jump-robust variation, or latent-volatility filters would provide alternative stress coordinates.

Terminal-universe survivorship. The 419 current identities are projected backward. Historical breadth and duration are conditional on this selected survivor panel.

Vendor vintage. Yahoo Finance does not expose a point-in-time archive of successive adjusted-price vintages. The database preserves the downloaded run, but a future download can differ. Immutable raw snapshots and hash-based run identification are required for reproducible production research.

Option-history limitation. Only currently listed option chains are discoverable. The option layer is a timestamped snapshot archive and is not evidence about historical leg membership. It is excluded from the classifier by design.

Multiple testing and cross-sectional dependence. The HAC p-values receive Benjamini-Hochberg adjustment, but the bootstrap, horizon, curvature, and tail filters do not share a unified multiplicity correction. Market and sector shocks also make ticker statistics cross-sectionally dependent.

Strict antifragility. Equation (10) measures the local first-order sign required by MASS. Taleb-style structural antifragility would require independently specified disorder shocks, convex payoff response over a domain, beneficial distributional perturbations, and reserved-sample replication. The exploratory filter is hypothesis generation only.

The appropriate falsification program is therefore to predeclare horizons and robustness rules; use longer histories or nonoverlapping observations for long-memory inference; reconstruct historical investable universes including delistings; compare daily, intraday, and jump-robust stress measures; preserve vendor vintages; multiplicity-correct exploratory tests; and evaluate the later density-pair operator on held-out periods. None of these extensions requires a third classification state.

10 Conclusion

The corrected result is unambiguous. The code is an exhaustive two-leg state classifier. On the fixed 11 July 2026 run, it assigns 267 tickers to the fragile response leg R^+ and 152 to the antifragile response leg R^- . All 419 current assignments are independently reproduced from the coefficient sign, and 3,439,101 historical classified rows contain zero sign-leg mismatch. Every valid ticker occupies one unique leg on one date.

The 120 strong classifications answer a different question. They are the subset that pass six additional robustness diagnostics; they are not the only correct assignments. Extended tests show that robustness is sensitive to block length and horizon choice, exactly why confidence must accompany the leg without replacing it.

The classifier provides the dated R^+ and R^- inputs required by the MASS pair operator. Pair density, stress activation, transition confirmation, and allocation belong downstream. Options are stored and causally aligned but do not participate in classification. No return prediction is used to construct the two legs.

Data and code availability

The empirical artifacts used for this article are the current classification CSV, the compressed daily-history CSV, the SQLite database, and the single Python pipeline in the project workspace. The run identifier and SHA-256 fingerprints in Appendix E fix the exact empirical vintage. Yahoo Finance data remain subject to the provider's terms and revision policy.

Declaration

This is a methodological study, not investment advice. The labels are time-varying statistical states and are not recommendations to buy, sell, or hold any security.

A Algebraic sign convention

For the conventional regression $Y = a + bX + e$,

$$\hat{b} = \frac{\sum_s (X_s - \bar{X})(Y_s - \bar{Y})}{\sum_s (X_s - \bar{X})^2} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}. \quad (13)$$

The MASS relation writes the slope as $-\beta$, so

$$\hat{\beta} = -\hat{b} = -\frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}. \quad (14)$$

If $\hat{\beta} > 0$, then $\hat{b} < 0$ and adjusted performance decreases with realized variance: R^+ . If $\hat{\beta} < 0$, then $\hat{b} > 0$: R^- . The independent-beta error in the export compares this direct reconstruction with the stored rolling coefficient. It is a software identity check, not independent economic evidence.

B Algorithmic specification

For each requested ticker:

1. Download Yahoo Finance OHLC and corporate-action fields; retain positive, unique adjusted closes ordered by market date and version every row by `run_id`.
2. Apply the ten-year, staleness, and weekday-coverage eligibility rules.
3. Compute daily log returns, 252-session annualized realized variance, and 252-session annual excess log performance.
4. On every date with sufficient valid history, compute 126-, 252-, and 504-row signed response coefficients.
5. Assign exactly one leg from the 252-row sign using Equation (10); store the next observed session as the effective date when it exists.
6. At the terminal date, reconstruct the coefficient independently and execute the six confidence diagnostics.
7. Store the exploratory curvature-tail filter separately for negative-sign research candidates.
8. Download all currently discoverable option expirations as timestamped snapshots, then align them only to OHLC and regime information available by the option cutoff. Do not feed options into the classifier.
9. Export all daily calculations and the complete current cross-section. Never replace the leg, coefficient, diagnostics, and timestamps with a single ambiguous validation flag.

C Current export contract

The 63-column current CSV can be read in the following functional blocks. Field names are set with breakable monospace text so they remain copyable without being clipped at a page margin.

Identity	<code>run_id</code> , <code>ticker</code> , <code>yahoo_symbol</code> : run vintage, requested identity, and provider symbol.
Eligibility	<code>first_price_date</code> , <code>last_price_date</code> , <code>history_years</code> , <code>staleness_days</code> , <code>coverage_ratio</code> , <code>price_basis</code> , <code>eligible_10y</code> : history eligibility and price-source audit. Coverage uses expected weekdays, not an exchange-holiday calendar.
Sample size	<code>observations</code> , <code>regression_observations</code> : available price observations and valid regression rows.
Full-history fit	<code>beta_full</code> , <code>alpha_full</code> , <code>r2_full</code> : descriptive full-history regression, which does not determine the current leg.
Current fit	<code>beta_rolling</code> , <code>alpha_rolling</code> , <code>r2_rolling</code> , <code>regime_date</code> , <code>regime_available_utc</code> , <code>regime</code> , <code>error</code> : current 252-row estimate, signal date, availability, leg alias, and error state.

Leg and horizons

`leg`, `beta_126`, `beta_252`, `beta_504`: unique R^+/R^- assignment and nested-horizon coefficients.

Exact reconstruction

`independent_beta_error`, `classification_rule_valid`: closed-form reconstruction error and exact sign-mapping audit.

HAC inference

`hac_beta`, `hac_se`, `hac_ci_low`, `hac_ci_high`, `hac_pvalue`, `hac_fdr_qv`, `hac_valid`: Newey-West inference and cross-sectional false-discovery adjustment.

Bootstrap

`bootstrap_sign_probability`, `bootstrap_ci_low`, `bootstrap_ci_high`, `bootstrap_valid`: circular moving-block bootstrap sign evidence.

Perturbations

`window_sign_consensus`, `delete_block_sign_stable`, `winsorized_sign_stable`: horizon, deletion, and outlier checks.

Persistence

`sign_consistency_63`, `sign_consistency_126`, `sign_consistency_252`: recent shares of daily rolling signs equal to the current sign.

Curvature

`convexity_coefficient`, `convexity_pvalue`, `jensen_convexity`: exploratory local quadratic-curvature fields.

Tails

`k_threshold_05`, `k_threshold_01`, `left_tail_semi_vega_05`, `left_tail_semi_vega_01`, `right_tail_semi_vega_95`, `right_tail_semi_vega_99`, `tail_asymmetry_05_95`, `tail_asymmetry_01_99`, `left_tail_robust`, `right_tail_beneficial`: exploratory conditional tail thresholds, semideviation differences, asymmetries, and flags.

Eligibility flags

`mass_validated`, `taleb_validated`, `pair_eligible`: respectively, correct eligible MASS sign rule; legacy-named exploratory curvature-tail candidate; and downstream pair-stage eligibility.

Confidence

`passed_checks`, `classification_confidence`: number of the six production diagnostics passed and the separate weak/medium/strong grade.

The daily export contains the raw and intermediate fields needed to reconstruct returns, rolling moments, coefficient paths, residuals, leg labels, sign persistence, signal dates, and next-observed-session effective dates for all 3,649,858 stored rows.

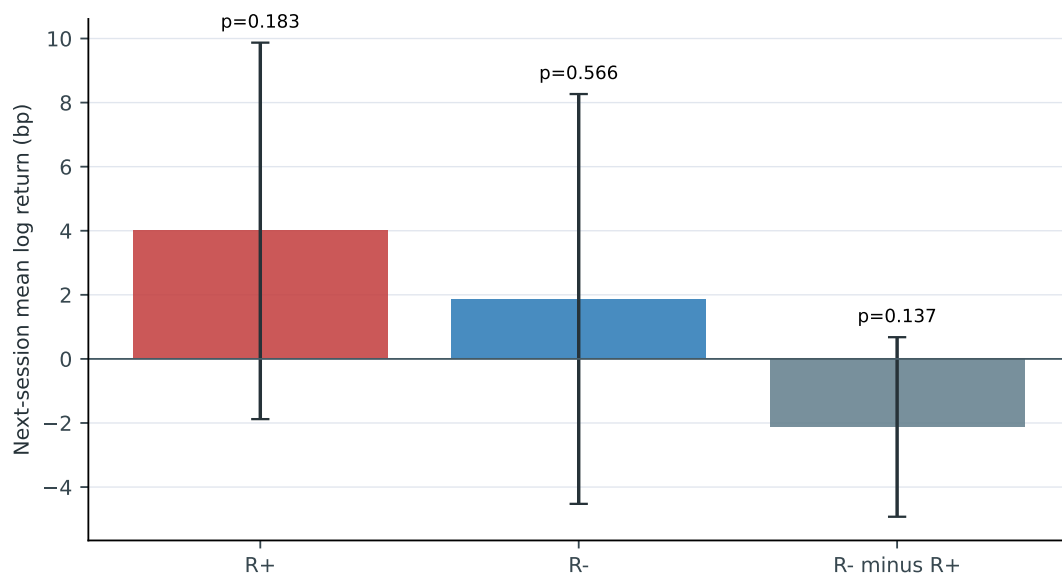
D Out-of-scope next-session diagnostic

This appendix is deliberately outside the classifier. It is not used to assign R^+/R^- , grade confidence, or select pairs. It is retained only to correct the earlier descriptive calculation.

The final signal date, 10 July 2026, has no downloaded following return. The balanced forward sample therefore contains 2,014 dates from 3 July 2018 through 9 July 2026. For each signal date, the equal-weight mean log return at the next observed session is computed within each leg. Time-series means receive 21-lag HAC covariance.

Table 6: Corrected out-of-scope next-session diagnostic. Returns and intervals are in basis points.

Series	Mean	95% HAC low	95% HAC high	p-value
R^+	4.00	-1.88	9.87	0.183
R^-	1.87	-4.53	8.27	0.566
R^- minus R^+	-2.12	-4.93	0.68	0.137

**Figure 7:** Corrected next-session descriptive means. The diagnostic is statistically insignificant and has no role in constructing the legs.

The result neither validates nor invalidates classification accuracy. It asks a separate prediction question, which is why it is excluded from the abstract, main results, and conclusion.

E Reproducibility fingerprints

Table 7: SHA-256 fingerprints of the exact empirical inputs. The strings use `\url` line breaking to prevent clipped or split-copy DOI/hash errors.

File	SHA-256
fragility_classification_results.csv	2aaff6ecbcacd22715b30b3cda3e978f0306055815e4b6c2caa7b0bc b62a5323
fragility_daily_history.csv.gz	f238d4c87ddad3ac54e1d3c0a8441cc52ea615f723ced66eac1b5293 63502b45
yahoo_fragility_pipeline.py	4a06cf2defa8caa7739e15302f6b42345b12612f4a7b9bb6ccd6e707 9e1dc050

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